

and Sn—exhibit similar properties as graphene and have no lacking while being used in digital electronics. Thus, 2D electronics, which got its origin in graphene, has moved into other materials like silicene, germanene, and phosphorene. These materials have additional advantages like excellent magnetic properties, tuneable band gap, active optical properties, and so on.

2D materials like silicene, germanene, and phosphorene have tremendous potential in the field of optoelectronics as light emitting diodes (LEDs), optical signal processors, and solar photovoltaic panels. Owing to their light weight and flexibility they are also suitable for flexible design of wearables and other flexible devices.

2D materials can be used to produce a transistor equivalent to the size of a molecule. Size and stability of these pave the way for their use in computing. The most promising application of 2D electronics is in solar energy industry, where any surface like paper, walls, windows, glass of consumer durables, and so on can be converted into solar cells. 2D electronics finds its use in supercapacitors, fuel cells, and more.

Organic electronics

In the year 2000, the Nobel Prize for Chemistry was given to Alan Heeger, Alan Macdiarmid, and Shirakawa for the development of conductive polymers. In common parlance, they proved that even plastic can conduct electricity. This paved the origin of organic electronics. Organic electronics materials are made of organic molecules (carbon based materials) or polymers by chemical synthesis. Organic electronics materials also include a variety of dyes and other organic molecules.

Organic materials are yet to compete in performance, if compared with conventional inorganic conductors and semiconductors, but

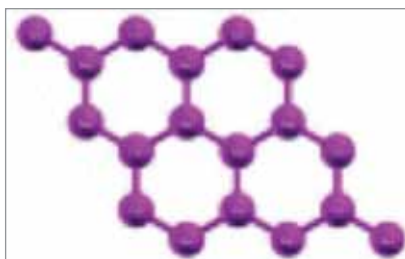


Fig. 1: Top view of optimised 3x3 supercell of monolayer germanene

they come with several advantages over the conventional ones. They have low material cost as well as production cost, have adaptivity of synthesis, are bio-compatible, and mechanically flexible.

We already are familiar with the organic LED (OLED) screens for our smartphones and TV sets. But their potential for future application stretches far beyond. Organic material enables us to fold our smartphones or other portable devices like a paper.

Healthcare systems will see new possibilities that organic materials offer, where they can interact with our biological cell and sensory organs. Healthcare systems can respond to signals at cellular level in our body with the capability added by organic electronics. Owing to this possibility, our artificial organs or limbs can work very near to the natural in terms of feeling and reaction. Artificial skin that is close to natural skin is not very far off.

Organic electronics unleashes the possibility of printing solar cells over plastic sheets that can be folded like a map, ready to be installed anywhere. As the organic electronics materials will be synthesised in the lab and will be bio degradable, they will have better sustainability cycle when compared to conventional materials, which have dependency on mined materials from the earth.

Memristors

The word memristor has been derived from the two words: memory

and resistor, meaning thereby that it is a resistor with memory. In theory, memristors were predicted as early as 1971 by Leon Chua and were explained through some mathematical equations. The predication said that we were missing an electronics component like memristor. But the device was physically produced much later in 2008 by Stanley Williams.

The resistance of a memristor can be programmed and at the same time the device memorises its value. Recent developments in memristors have made them very reliable as they can be cycled more than 100 billion times and can remain stable for more than 10 years.

We are very much familiar with random access memory (RAM) and flash memory (such as in pen drives). In RAM, memory mobility is very high, but the disadvantage is that it does not retain the data in power-off mode. Flash drive retains the data in power-off condition, but its memory mobility is very slow. Memristor combines the capability of both, as it has very high mobility and also retains the data even in the power-off mode.

Memristor, when compared with a conventional solid-state device, requires less energy and has a faster speed of operation. At the same time, memristor can accommodate double the data in same physical size of device compared to a conventional semiconductor. Not only that, but our conventional semiconductors, when exposed to radiation, either lose the data or the data is distorted, whereas memristors are immune to the radiation.

In our conventional computing systems, data is stored in a location (memory) and processed in another location (processor). The data traffic between these two locations reduces the speed of computation and at the same time consumes huge amount of energy. Memristors store